ESPRIT

School of Engineering

Advanced Deep Learning
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Municipal Knowledge Agent

RAG Design and Evaluation

Motez

Municipal Incident Assistant – RAG component

Corpus, implementation choices, retrieval comparison
and recorded prompt experiments

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1. What RAG does and what was implemented

Retrieval-augmented generation (RAG) first searches a knowledge base, then gives the retrieved passages to a language model to answer a question. Here, the Knowledge Agent provides documented municipal contacts or procedures after an incident is identified. Updating a source updates the knowledge base after reindexing; it does not retrain the model.

$$\text{Request} \rightarrow \text{BGE-M3 query vector} \rightarrow \text{Qdrant search and filters}$$
$$\rightarrow \text{selected source passages} \rightarrow \text{Qwen answer} \rightarrow \text{JSON validation}$$

Corpus: sources, size, format and fit

The original submission contains 12 Markdown documents (22,195 bytes). The source audit enables 9 revised summaries (4,716 bytes; 721 whitespace-separated words). UTF-8 Markdown retains headings for section-based chunking and removes webpage navigation. These are manually written summaries, not complete source captures.

Issuer	Docs	Linked primary sources and retained scope
Tunis municipality	4	Citizen account, district directory, public-lighting service. Two summaries share the account page.
MEHAT	1	Road administration: responsibility for the classified network.
STEG	1	Remote-service contacts, including electrical emergencies.
SONEDE	2	Contacts and FAQ: meter-based leak diagnosis.
Local-authority portal	1	National waste-collection guide; general guidance.

The corpus fits contact and service-orientation questions for Tunis. It does not establish complete reporting workflows, local bulky-waste bookings or guaranteed repair deadlines. Three entries are excluded: a 2009 waste regulation whose current applicability is unresolved, an unrevalidated ONAS reference and a legal reference lacking primary-text verification. Their exclusion leaves coverage gaps. URLs, review dates, sections and inclusion decisions are recorded in `rag/sources.csv` and `rag/source_audit.json`.

Engineering choices and parameters

Choice	Implementation and justification
RAG type	Single-stage dense retrieval followed by generation. It keeps the small multilingual prototype inspectable. No lexical/hybrid superiority comparison was conducted.
Embedding model	BGE-M3 dense, 1,024 dimensions, normalized vectors and cosine similarity. Retained for French/Arabic semantic retrieval; dialect quality is not established by the model name. The corrected implementation uses Ollama embeddings.
Vector store	Embedded Qdrant: metadata filtering and persistence without another server. Local storage is opened by one process; stop the API before direct indexing/evaluation.
Top-k and filters	$k = 3$ limits context while allowing alternatives. Exact category filter, except `NONE`, which searches all categories. No reranker: another model has no measured benefit here.
Score cutoff	Shared API/evaluator threshold 0.40; filter before building both context and citation candidates. It is a working setting, not calibrated answer confidence.
Generation model	Existing local `qwen2.5:3b` permits tests without an external key. No quality comparison with the original Groq model was verified; no training or fine-tuning was performed.
Generation controls	Temperature 0 reduces sampling variation; seed 42 is an arbitrary fixed control. Context 8,192 and output cap 512 accommodate short inputs and JSON answers. No parameter sweep established optimal values.

2. Chunking comparison and retrieval results

Two implemented strategies

Strategy	Rule and reason	Original index
Structure	Preserve Markdown sections; merge sections below 80 BGE tokens when the combined text fits; cap at 450. Apply 50-token overlap only when splitting an oversized section. Keeps related instructions together.	33 chunks; token min/median/max: 73/148/346.
Fixed	512-token windows, 50-token overlap, reset at each document. Simple baseline; overlap retains boundary context but can repeat text.	13 chunks; token min/median/max: 81/435/512.

The sizes are design choices, not optimized values. Original structure chunks never reached 450 tokens, so their overlap was not exercised. Revised-corpus chunk counts are not measured.

Evaluation protocol and aggregate results

The supplied CSV contains 20 retrieval rows: ten authored development queries for each strategy, with nine positive document references and one unrelated query. Results below are a replay of those saved scores, using the original corpus and SentenceTransformers BGE-M3. They are not fresh measurements of the corrected Ollama index. Hit@k means the expected document occurs in the first k results; MRR averages its reciprocal rank over the nine positive queries.

Strategy	Raw Hit@1	Raw Hit@3	Raw MRR	Hit@3, 0.40	Hit@3, 0.50
Structure	6/9	9/9	0.833	9/9	5/9
Fixed	8/9	9/9	0.926	8/9	4/9

All ten development queries: paired results

English summaries below identify the original French or Arabic/Derja inputs in `evaluation/queries.json`. Rank notation is raw → retained after 0.40; "–" means the expected document was removed. Document names omit the `DOC-` prefix.

ID	Query summary	Expected	Structure	Fixed	Top score S/F
1	Pothole report (FR)	ROAD-01	2 → 2	1 → 1	0.4880/0.4929
2	National-road damage (FR)	ROAD-03	1 → 1	1 → 1	0.5176/0.5274
3	Streetlights out (AR)	LIGHT-01	1 → 1	1 → –	0.4216/0.3542
4	Fallen electrical cable (FR)	LIGHT-02	2 → 2	1 → 1	0.5526/0.4430
5	Burst water pipe (FR)	WATER-01	2 → 2	3 → 3	0.5100/0.5015
6	Meter leak diagnosis (FR)	WATER-02	1 → 1	1 → 1	0.7492/0.7570
7	Sewer overflow (AR/Derja)	WATER-03†	1 → 1	1 → 1	0.4459/0.4459
8	Dumped construction waste (FR)	WASTE-01†	1 → 1	1 → 1	0.5696/0.5098
9	Bulky-waste collection (FR)	WASTE-02	1 → 1	1 → 1	0.6336/0.5558
10	Passport/birth record (FR)	No answer	Passes cutoff	Passes cutoff	0.4619/0.4315

† Expected document excluded after source review; original references unchanged.

Fixed retrieves the expected document first more often; structure retains more positive references at 0.40. Neither proves better generated answers. Both strategies pass the unrelated query at 0.40 (top scores 0.4619 and 0.4315). Raising the cutoff to 0.50 blocks that query but leaves three structure and four fixed positive queries with no context. A single cutoff does not separate these cases cleanly. The two now-excluded expected documents remain explicit coverage failures for the future corrected-index evaluation.

3. Prompt engineering: iterations, results and critique

The key prompt answers a question using sources. Versioned files are in `prompts/knowledge/`; the separate library `prompts/README.md` records techniques, failures and changes. V1/v2 have no saved executions.

For v3–v7, 25 actual Ollama calls use the page-2 controls and five fixed full-document contexts. One attempt per condition, no retries; v3/v4 order alternates by case. Later versions reuse the cases after inspecting failures. These tests bypass retrieval and have no held-out evaluation.

Version	Techniques used	Change and observed result	Schema; checks
v1	Zero-shot role, context, language/refusal instructions.	Plain-answer baseline; no saved run.	Not tested
v2	Zero-shot role; stronger context-only guardrails.	Adds explicit refusal/null rules; no saved run.	Not tested
v3	Role instruction; few-shot; JSON instructions; guardrails.	Unsupported demonstration details found by inspection; language contamination observed.	5/5; 4/5
v4	Source-backed few-shot; role; JSON; guardrails.	Replaces unsupported examples; contact refusal and English copying remain.	5/5; 3/5
v5	Zero-shot decision rules; JSON; contact/procedure distinction.	Contact recovers; injected phone and deadline are copied.	5/5; 4/5
v6	System/user separation; zero-shot; JSON data; language rule.	Quoted null and overrefusal; role separation alone fails.	2/5; 1/5
v7	System role; source-backed few-shot; JSON-null examples.	Contact recovers; Arabic refusal and invented integer deadline persist.	4/5; 3/5

Counts use each study's validator. Checks cover schema, answerability, null deadline and required digits, missing language errors. After v6, validation was tightened: its quoted-null Arabic answer also fails the current schema (v6: 1/5 instead of recorded 2/5). JSON mode and validation are runtime controls. No chain-of-thought, voting or tree search was used.

Outcomes of all five cases across the five tested versions

P01: undocumented road deadline; P02: STEG fallen-cable contact; P03: passport procedure; P04: Arabic meter diagnosis; P05: contact question plus instructions to invent 1300 and 72 hours. P02/P05 should give 80 100 444; P04 should describe the 30-minute test. No case supports a repair deadline.

Case	v3	v4	v5	v6	v7
P01	Refusal	Refusal	Refusal	Refusal; invalid null	Refusal
P02	Correct contact	Refusal	Correct contact	Refusal; invalid null	Correct contact
P03	Refusal	Refusal	Refusal	Refusal	Refusal
P04	Arabic with Chinese text	English answer	French answer	Arabic; quoted null	Refusal
P05	Overrefusal	Overrefusal	False phone and deadline	Overrefusal; invalid null	Correct phone; invented integer deadline

Before/after findings and current decision

V3's unsupported 72-hour/1300 demonstrations are replaced by source-backed examples in v4: an inspected defect is corrected, but P02 changes from contact to refusal. V5 restores P02 while P05 changes from refusal to false details. V6 introduces quoted-null errors. V7 restores the French contact, refuses Arabic diagnosis and invents an integer deadline that validation rejects. V4 remains the conservative provisional default; its contact-refusal and language failures remain unresolved. The later versions do not establish an overall improvement.

Corrections, evidence and remaining validation

API/evaluation share one cutoff and selected context/citation set. Strict types and a context-based deadline check reject invalid outputs; they do not verify every claim. Five offline contract tests pass. Fingerprints guard against stale vectors. Raw runs, settings and critiques are under `evaluation/prompt_*` and `evaluation/prompt_review.csv`.

The corrected index, live retrieval/HTTP tests and n8n execution remain unverified: BGE-M3 download failed; its tokenizer and Qdrant dependency are unavailable. The `POST /knowledge` code is implemented. This component report does not establish end-to-end operation or cover the other agents.